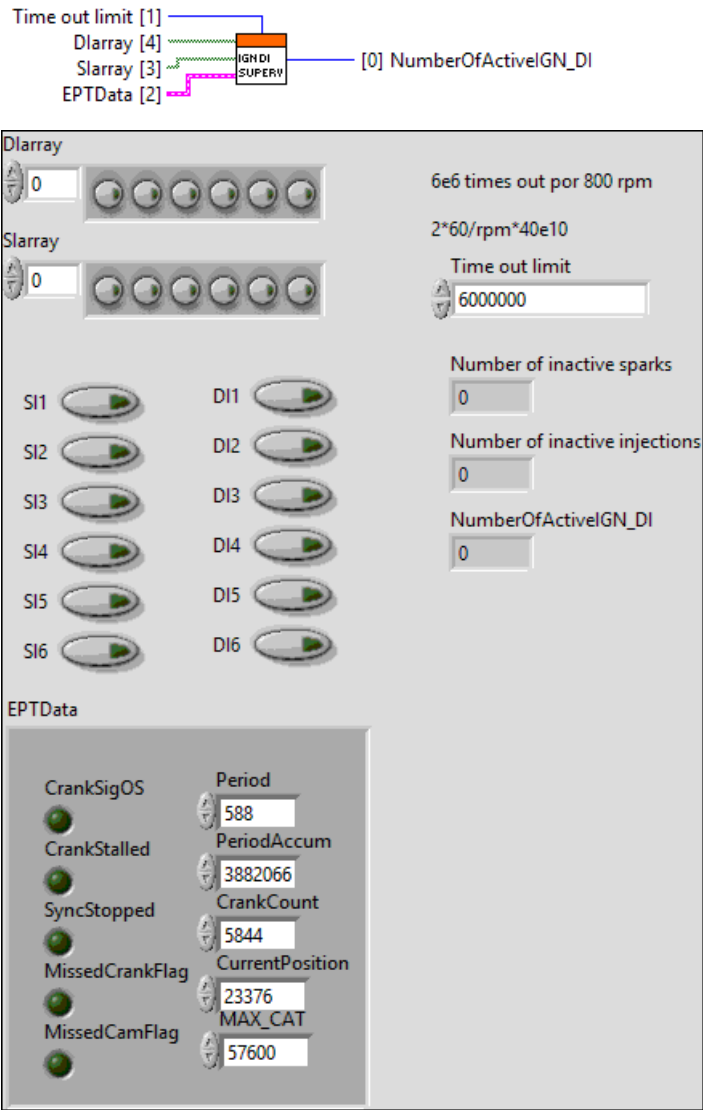


APC\_9049\_FPGA\_IGNDI\_supervisor.vi



EPTData

Cluster of status information related to engine position tracking.

CrankSigOS

A 40 MHz one-clock one-shot generated at the rising edge of each received or simulated CrankSig input. This signal can be used in conjunction with CrankCount for any purpose within LabVIEW FPGA logic to perform engine synchronous tasks.

CrankStalled

Set to TRUE when the engine speed is below stall speed.

SyncStopped

Set to TRUE while position tracking is not taking place. In other words, no sync.

MissedCrankFlag

Set to TRUE upon the detection of a CamSig rising edge while the CrankCount value has not yet reached the expected value. In other words, the Cam tooth has been detected sooner than expected. This condition causes a loss of sync. Re-sync is not allowed until the flag is cleared by setting MissedCrankFlagClr to TRUE.

MissedCamFlag

Set to TRUE upon CrankCount being incremented past the expected value for the location of a CamSig rising edge before the CamSig rising edge is detected. In other words, the cam tooth was not received when expected. This condition causes a loss of sync. Re-sync is not allowed until the flag is cleared by setting MissedCamFlagClr to TRUE.

Period

The period between the last two crank trigger teeth, as if there were no missing or extra teeth, in terms of 40 MHz clock ticks. Period will be set to 0 while Stalled is TRUE. Period in clock ticks can be converted to instantaneous engine speed in RPM according to the following equation:

$$\text{EngineSpeedInstant(RPM)} = (60 * 4E7) / (\text{Period} * \text{NumberOfCrankTeeth})$$

PeriodAccum

The accumulated period over PeriodAccumCount crank teeth in terms of 40 MHz clock ticks. PeriodAccum will be set to 0 while Stalled is TRUE. PeriodAccum in clock ticks can be converted to average engine speed in RPM according to the following equation:

$$\text{EngineSpeedAve(RPM)} = (60 * 4E7) / (\text{PeriodAccum} * \text{NumberOfCrankTeeth} / \text{PeriodAccumCount})$$

CrankCount

The latest tooth count referenced to crank tooth 0. Tooth 0 is determined according to the pattern that is being tracked and is discussed within the pattern specific documentation. CrankCount is provided in terms of a complete engine cycle and is reported as if there were no missing or extra teeth in the pattern. For example, if PATTERN = N-M, NumberOfCrankTeeth = 60 and Stroke = TRUE, then CrankCount will range from 0 to 119.

CurrentPosition

During sync, the EPT provides the latest crankshaft position in terms of absolute Crank Angle Ticks (CAT) over a complete engine cycle, referenced to position 0. Position 0 is determined according to the pattern that is being tracked and is discussed within the Pattern-Specific Information section at the end of the EPT user manual. The CurrentPosition in CAD can be calculated according to the following:

$$\text{CurrentPosition(CAD)} = [\text{CurrentPosition(CAT)} * \text{Stroke} * 360] / \text{MAX\_CAT}$$

Where Stroke = 2 if 4-stroke and Stroke = 1 if 2-stroke.

MAX\_CAT

Maximum Crank Angle Ticks per engine cycle.

A Crank Angle Tick (CAT) is a single unit of angular measure reported by the CurrentPosition output of the EPT VI. Reported as a power-of-two angular ticks of crank position travel, having a resolution dependent on EXTRAP and the number of physical teeth per crankshaft rotation. For example, if using the N-M EPT VI, which has an extrapolation of 7, the number of CAT per crank tooth would be 2^7=128, and CurrentPosition would be evenly incremented by 128 CAT from one physical tooth to the next. If a 60-2 pattern were used, the maximum number of CAT per crankshaft rotation (cycle) would be 60\*128=7680. If the engine was a 4-stroke, the total number of CAT per engine cycle would be 2\*60\*128=15360.

Time out limit

Dlarray

DI1 2

Slarray

Slarray

NumberOfActiveIGN\_DI

